

# Hein Zarni Aung

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## EDUCATION

### Stony Brook University

Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Full Stack Development, Software Security, Systems Fundamentals I & II, Algorithms, Statistics & Probability

## RESEARCH

### Undergraduate Research Assistant — Ethos Lab, SBU

Sep 2025 – Dec 2025 (paused)

PI: Abisheka Pitumpe | Team size: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites; auto-captured homepage screenshots of promising candidates to speed up manual review.
- Collaborated in a two-person team with one honors student under a faculty PI to design keyword- and heuristic-based detection logic for scam website identification.

## INTERNSHIPS

### Software Engineering Intern — Clustr

Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Owned end-to-end delivery across an ambiguous, multi-part build spanning backend and frontend — server infrastructure, a scheduling module, and authentication — contributing 50+ commits alongside an 8-person dev team.
- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.
- Designed and implemented a Scheduler UI module featuring tabbed navigation, a reschedule bottom sheet, custom date/time picker integration, and a calendar-completion animation to improve user engagement.

### Software Engineering Intern — Prenora Dynamics LLC

Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, researching and evaluating SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).

## PROJECTS

### TuneAcademy — Hack Brooklyn 2026 (36-hour hackathon)

Apr 2026

Winner: Best Prenora Dynamics Hack (Sponsor Challenge) | Team of 4

- Built the AI/audio-analysis pipeline for an app matching music students with instructors, using Essentia and NumPy to extract pitch, rhythm, and tone features from recorded performances.
- Took the initiative to pivot the team from a planned ML model to rule-based scoring algorithms under hackathon time and compute constraints, delivering reliable weakness detection within the 36-hour build window.

## CAMPUS INVOLVEMENT

### Teaching Assistant – Data Structures & Algorithms

Aug 2025 – Dec 2025

Dept. of Computer Science, Stony Brook University

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures (linked lists, trees, heaps, hash tables, graphs).
- Created original review materials from scratch for midterm exam preparation; materials were adopted as the basis for subsequent semesters' review sessions.

### International Peer Mentor — Office of Global Affairs, SBU

Aug 2025 – Present

- Mentor 6 international students through the university's peer mentoring program, providing biweekly guidance on academic and cultural adjustment.

## SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C

**Frontend:** React, React Native, Next.js

**Backend:** Node.js, Express.js, FastAPI

**Databases & Cloud:** PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP

**Tools:** Git, Docker, Linux, PM2, Selenium